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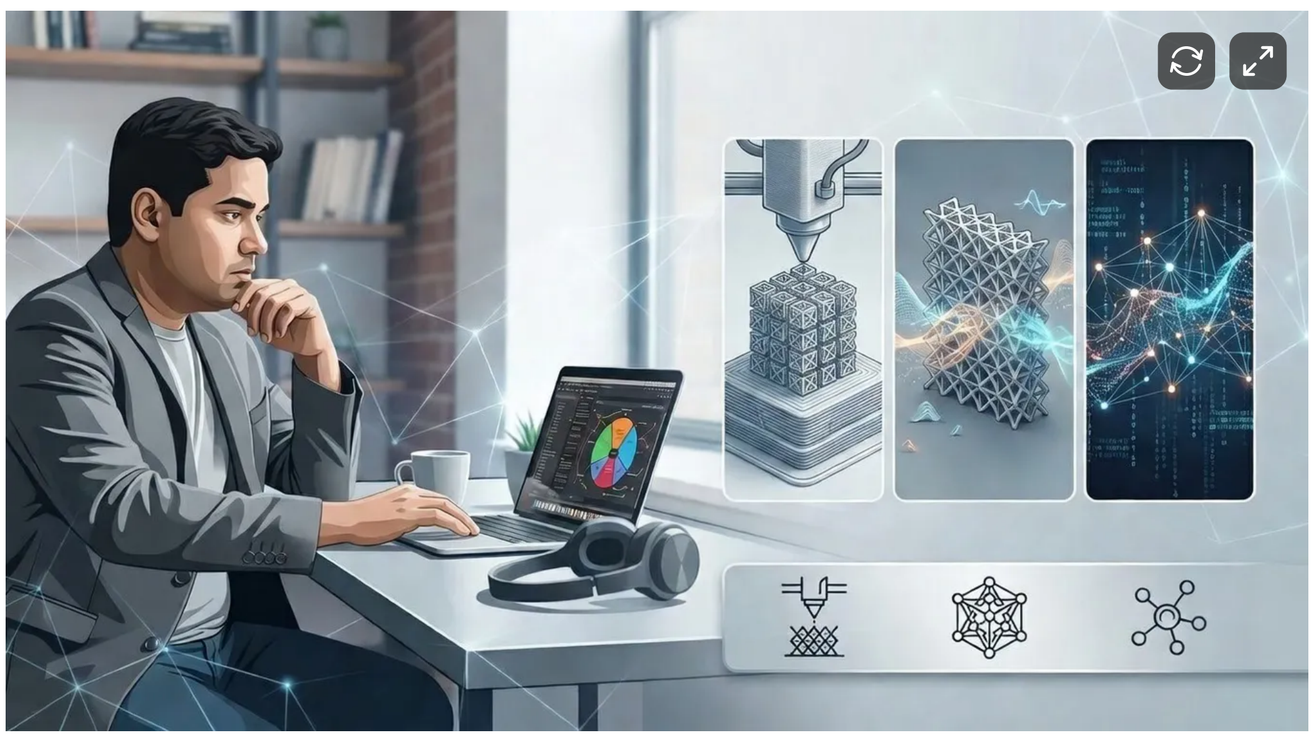

Shahadat Hussain, PhD

Researcher in 3D printing, metamaterials and shape memory alloys.

Engineering the Future, One Lattice at a Time

Dr. Shahadat Hussain's platform for unfiltered insights into additive manufacturing, metamaterials, shape memory alloys, and data science applications. Striving for rigorous, transparent research — from lab failures to breakthroughs. A hub for engineers, researchers, and innovators.

The Mission

I am **Dr. Shahadat Hussain**, a researcher and engineer dedicated to the intersection of materials science, advanced manufacturing, and computational design.

My work is driven by a central question: *How can we design internal structures to give materials extraordinary properties?* From 3D-printed shape memory alloys to complex TPMS (Triply Periodic Minimal Surface) lattices, I explore the frontier of what is physically possible through additive manufacturing.

My Background

Currently, I serve as a **Research Associate at Khalifa University** in Abu Dhabi. My journey in science has taken me from a PhD in Materials Science and Technology (AcSIR, India) to deep-diving into the mechanics of "smart" materials.

Over the years, I've developed a specialized toolkit that includes:

- **Additive Manufacturing:** Mastering Laser Powder Bed Fusion (LPBF) and FFF.
- **Smart Materials:** Exploring the "memory" of NiTi and Cu-based alloys.
- **Computational Design:** Using Python, SQL, and Data Science to predict how materials will behave before they are even printed.

Why I Started This Substack

The world of materials science can often feel locked behind paywalls and dense academic jargon. I created this space to pull back the curtain on:

1. **Technical Deep-Dives:** Breaking down the latest trends in metamaterials and 3D printing.
2. **Research Insights:** Sharing the "ah-ha" moments (and the failures) from my lab work at Khalifa University.
3. **The Digital Lab:** Discussing how Data Science and Scientific Computing are changing the way we do engineering.

Stay Connected

Whether you are a fellow researcher, a student, or someone fascinated by the future of manufacturing, I'm glad you're here. By subscribing, you'll get my latest insights and research updates delivered directly to your inbox.

Let's build something extraordinary.

People

Shahadat Hussain, PhD
@drshahadathussain

Researcher at the crossroads of atoms and bits. I explore the future of 3D printing and advanced materials through the lens of data science, testings, and scientific computing—bridging the gap between physical experiments and digital insights.

Shahadat Hussain, PhD

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